

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Integration

*Pure 2 integration mixes exact area, transformed estimates,
and curve geometry.*

INTEGRATION · P2 · 12

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 POWER: Integrate Powers

TRIGGER *integrate algebraic powers*

02 AREA: Area Between Curves

TRIGGER *area bounded by curves*

03 TRAP: Trapezium Estimate

TRIGGER *table of values, approximate area*

04 SHIFT: Transform Integral

TRIGGER *estimate related integral*

BANK EVIDENCE Local pages: 43 problem, 58 answer, 39 notes. Website primary entries: 29.

01 POWER: Integrate Powers

TRIGGER *integrate algebraic powers*

BECOMES Rewrite then apply the power rule.

FIRST LINE TO WRITE

$$\int ax^n dx = \frac{a}{n+1}x^{n+1} + c$$

SIMPLEST STRATEGY

- 1 Put roots/fractions as powers.
- 2 One added to power.
- 3 Write coefficient over new power.
- 4 Evaluate limits or add c .
- 5 Rewrite exactly.

WORKED MODEL

$$\int 6x^2 dx = 2x^3 + c.$$

HARD VARIANTS

- 1 Do not use the simple power rule for x^{-1} .
- 2 Simplify products or fractions before integrating.
- 3 Keep exact fractional powers until the final answer.

— BOTTOM LINE

Most Pure 2 integrals begin by rewriting powers.

02 AREA: Area Between Curves

TRIGGER *area bounded by curves*

BECOMES Find intersections, then integrate top minus bottom.

FIRST LINE TO WRITE

$$\text{Area} = \int_a^b (\text{top} - \text{bottom}) dx$$

SIMPLEST STRATEGY

- 1 Assign intersections.
- 2 Read top curve.
- 3 Evaluate the integral.
- 4 Answer positive area.

WORKED MODEL

$$\int_1^3 [(2x + 1) - x^2] dx.$$

HARD VARIANTS

- 1 If curves cross more than once, split the area.
- 2 Use the graph or test point to decide top curve.
- 3 Check restrictions such as $x > 0$ before choosing limits.

— BOTTOM LINE

Area is positive geometry, not signed displacement.

03 TRAP: Trapezium Estimate

TRIGGER *table of values, approximate area*

BECOMES Use width times end-plus-double-middle structure.

FIRST LINE TO WRITE

$$\frac{h}{2} \{y_0 + y_n + 2(y_1 + \cdots + y_{n-1})\}$$

SIMPLEST STRATEGY

- 1 Take the strip width.
- 2 Record end ordinates once.
- 3 Add middle ordinates twice.
- 4 Present rounded estimate.

WORKED MODEL

$$\frac{2}{2} \{1 + 5 + 2(2 + 3)\} = 16.$$

HARD VARIANTS

- 1 If a table value is unknown, set the trapezium formula equal to the given estimate.
- 2 Use the same estimate to transform related integrals.
- 3 State overestimate or underestimate from curve shape if asked.

— BOTTOM LINE

The middle ordinates are doubled.

04 SHIFT: Transform Integral

TRIGGER *estimate related integral*

BECOMES Rewrite the new integrand in terms of the old one.

FIRST LINE TO WRITE

$$\int (af(x) + g(x))dx = a \int f(x)dx + \int g(x)dx$$

SIMPLEST STRATEGY

- 1 Spot the old integrand.
- 2 Handle multiplier or shift.
- 3 Integrate extra simple part.
- 4 Fold in known estimate.
- 5 Ttidy units/rounding.

WORKED MODEL

$$\int (2f(x) - 3)dx = 2 \int f(x)dx - 3(b - a).$$

HARD VARIANTS

- 1 For $f(x + c)$ or $f(-x)$, check the limits by substitution.
- 2 Do not round the known estimate too early.
- 3 If the integrand includes x , integrate that exact extra part.

— BOTTOM LINE

Related estimates come from algebra before integration.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
integrate algebraic powers	$\int ax^n dx = \frac{a}{n+1}x^{n+1} + c$
area bounded by curves	$\text{Area} = \int_a^b (\text{top} - \text{bottom}) dx$
table of values, approximate area	$\frac{h}{2}\{y_0 + y_n + 2(y_1 + \cdots + y_{n-1})\}$
estimate related integral	$\int (af(x) + g(x))dx = a \int f(x)dx + \int g(x)dx$